

CSE Computation Cluster

User Manual

Cluster: csecluster
Login Node: 172.16.112.202
Version 1.0

*For internal users
Computer Science and Engineering
IIT Guwahati*

1. Introduction

This manual describes how to use the CSE Computation cluster, which is managed by SLURM (Simple Linux Utility for Resource Management). SLURM is an open-source workload manager responsible for scheduling and running jobs across the compute nodes. All submitted jobs run through SLURM to ensure fair resource sharing among users.

Before submitting jobs, users must connect to the login node, prepare their job scripts, and select the appropriate partition based on their resource requirements.

1.1 Cluster Overview

Parameter	Value
Cluster Name	csecluster
Login Node	clusterlogin / 172.16.112.202
Scheduler	SLURM (Backfill)
Resource Selection	Conservative Resources (Core + Memory)
GPU Types Supported	NVIDIA H100 NVL, Tesla P100-PCIE-16GB

2. Compute Nodes

The cluster has three nodes: one login node/ two compute node and 7 GPU nodes.

Node Name	CPUs	Memory (GB)	GPU	GPU Model
Clusterlogin (172.16.112.202)	32	256	None	—
cpu-01	32	256	None	--
gpu-h100	48 (2s x 12c x 2t)	256	1 x GPU	NVIDIA H100 NVL
gpu-P100-01	32 (2s x 8c x 2t)	128	2 x GPU	Tesla P100-16GB
gpu-P100-02	32 (2s x 8c x 2t)	128	2 x GPU	Tesla P100-16GB

Note: The clusterlogin node is shared. Avoid running heavy computation directly on this node without submitting a SLURM job.

3. Partitions

Partitions define which nodes are available and which user groups are allowed. Choose the correct partition when submitting a job.

Partition	Node	Default?	Max Time	Allowed Accounts
cpu	cpu-01	YES	Unlimited	All users
gpu-H100	gpu-h100	NO	Unlimited	faculty, officer, phd
gpu-P100	gpu-16GB-01 gpu-16GB-02	NO	Unlimited	faculty, officer, phd, mtech, btech

4. Home Directory Quotas

Quota Type	Limit	Description
Soft Limit	50 GB	You can temporarily exceed this, but must reduce within 15 Days
Hard Limit	200 GB	Absolute maximum - you cannot write beyond this limit

● **CRITICAL:** The Most Important Rule : Before running a job please clear your old files from the home directory, as the new job result will be write to your home directory. If there is no space left your job will failed.

5. Connecting the Cluster

5.1 Login to the Cluster

User must connect to the login node(clusterlogin) for submitting any job.

```
ssh your_username@172.16.112.202
```

For linux use system terminal to ssh to the cluster.

For Windows: use putty/mobaxterm or any other ssh terminal.

5.2 Change Your Password (First Login)

```
$ passwd
```

5.3 Check Available Resources

```
##See all partitions and nodes

$ sinfo

##Detailed node information

$ sinfo -o "%P %N %c %m %G %a"

##Check specific partition

$sinfo -p gpu-P100

##Check your active job

$squeue -u $USER
```

6. Submitting Jobs

SLURM (Simple Linux Utility for Resource Management) is a workload manager that provides a framework for job queues, allocation of compute nodes, and the start and execution of jobs. User must use a bash script(slurm script) to request a particular computing resource and submit the job.

```
$sbatch my_job.sh
```

● THE MOST IMPORTANT RULE (DO NOT FORGET!)

Always start your job script with `cd "$TMPDIR" || exit 1`

This is the **#1 reason** jobs run slowly. Without this line, your job runs on slow network storage.

Always write your output to `./results` directory

6.1 Sample Job Script (my_job.sh)

```
#!/bin/bash
#SBATCH --job-name=my_job      # Job name
#SBATCH --output=%j.out       # Output file (%j = job ID)
```

```

#SBATCH --error=%j.err          # Error file
#SBATCH --time=02:00:00         # Max runtime (HH:MM:SS)
#SBATCH --partition=gpu-P100    # Partition to use
#SBATCH --gres=gpu:1           # Number of GPUs
#SBATCH --cpus-per-task=8       # CPU cores
#SBATCH --mem=32G               # Memory

# === CRITICAL: Change to local SSD (MUST DO!) ===
cd "$TMPDIR" || exit 1

echo "Job $SLURM_JOB_ID running on $(hostname)"
echo "Working directory: $(pwd)"
echo "TMPDIR: $TMPDIR"
echo "Results will be saved to: ./results/"

# === YOUR COMMANDS HERE ===
# Write ALL output to ./results/ directory
python train.py --output ./results/

echo "Job completed at $(date)"

```

Job Script Directives Reference

Directive	Purpose	Example
--job-name	Job name	--job-name=training
--output	Stdout file	--output=%j.out
--error	Stderr file	--error=%j.err
--time	Max runtime	--time=04:00:00
--partition	Which partition	--partition=gpu-P100
--gres=gpu:N	Number of GPUs	--gres=gpu:2
--cpus-per-task	CPU cores	--cpus-per-task=8
--mem	Memory	--mem=32G
--nodes	Number of nodes	--nodes=1

6.2 Basic CPU Job

Use sbatch to submit a batch script. Create a file (e.g., myjob.sh) with the following contents:

```

#!/bin/bash
#SBATCH --job-name=my_cpu_job
#SBATCH --partition=cpu
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4

```

```
#SBATCH --mem=8G
#SBATCH --time=02:00:00
#SBATCH --output=job_%j.out

cd "$TMPDIR" || exit 1
echo "Running on node: $(hostname) "
your_program
```

6.3 GPU Job (H100)

To use the NVIDIA H100 GPU node, specify the gpu-H100 partition and request a GPU resource:

```
#!/bin/bash
#SBATCH --job-name=h100_job
#SBATCH --partition=gpu-H100
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --gres=gpu:1
#SBATCH --time=04:00:00
#SBATCH --output=h100_%j.out

cd "$TMPDIR" || exit 1
nvidia-smi
python train.py
```

6.4 GPU Job (P100, dual GPU)

The gpu-16GB-01 node has two P100 GPUs. You may request one or both:

```
#SBATCH --partition=gpu-P100
#SBATCH --gres=gpu:2          # Request both GPUs
```

6.5 Interactive Job

To get an interactive shell on a compute node:

```
srun --partition=cpu --ntasks=1 --cpus-per-task=4 --mem=8G --pty bash
```

7. Saving Results

After your job is successfully executed on the compute/gpu node, it will automatically copy back to your home directory. **./job_results/<jobid>**

8. Job Limitations

A user can only submit maximum 2 jobs at a time and only one job will be executed at a time.

9. Monitoring Jobs

Command	Description
squeue	List all jobs currently in the queue
squeue -u \$USER	List only your jobs
squeue -p gpu-H100	List jobs on a specific partition
scontrol show job <jobid>	Detailed info about a specific job
sacct -j <jobid>	Accounting info after job completes
scancel <jobid>	Cancel a running or pending job
sinfo	Show node and partition status
sinfo -p gpu-P100	Show status of a specific partition

9.1 Job States

State Code	Meaning
PD	Pending — waiting for resources
R	Running
CG	Completing — cleaning up after job
F	Failed
CA	Cancelled by user or admin
TO	Timed out
CD	Completed successfully

10. Account Management & Access

Your SLURM account determines which partitions you can use. The table below shows the mapping:

Account Type	Accessible Partitions
All users (default)	cpu
btech	cpu, gpu-P100
mtech	cpu, gpu-P100,gpu-A100
phd	cpu, gpu-P100, gpu-A100 gpu-H100
officer	cpu, gpu-P100, gpu-A100, gpu-H100
faculty	cpu, gpu-P100, gpu-A100, gpu-H100

To check your current SLURM accounts:

```
| sacctmgr show user $USER withassoc
```

Contact: If you need access to a GPU partition, contact your system administrator with your username and account type.

11. Troubleshooting Common Issues

Error / Symptom	Likely Cause	Solution
Job stuck in PD state	No resources available or partition limit reached	Check queue, reduce resource request, or wait
sbatch: error: Invalid account	Account not authorized for partition	Use an allowed account or contact admin
Job killed immediately	Exceeded memory or time limit	Increase --mem or --time
CUDA not available	GPU not requested or wrong partition	Add --gres=gpu:1 and correct --partition
Results not in ~/job_results/	Files not written to results/ in scratch	Write outputs to \$TMPDIR/results/
Cannot write to scratch	Wrong path or directory not created	Use \$TMPDIR variable, not hardcoded path

11. Quick Reference Card

Task	Command
Submit a job	<code>sbatch myjob.sh</code>
Submit interactive job	<code>srun --pty bash -p cpu --mem=4G</code>
Check queue	<code>squeue -u \$USER</code>
Cancel a job	<code>scancel <jobid></code>
Job details	<code>scontrol show job <jobid></code>
Past job stats	<code>sacct -j <jobid> --format=JobID,State,Elapsed,MaxRSS</code>
Node/partition status	<code>sinfo</code>
Check my accounts	<code>sacctmgr show user \$USER withassoc</code>
Monitor GPU	<code>nvidia-smi</code> (inside GPU job)
Where are my results?	<code>ls ~/job_results/<jobid>/</code>

For further assistance, contact your HPC system administrator.